WORKED SOLUTION

STEP I 2002, Question 13

Probability and Statistics • Conditional Expectation

STEP I 2002, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q13

The key idea

There are only six possible pairs (U, V), so everything comes down to a small table of joint probabilities and a little algebra on each case. For the quadratic, the sign of V decides everything, because U^2 is at most 1. For the cubic, the coefficients are suspiciously tangled up with U and V; that is a strong hint that it factorises, and once it does, the roots can be read off.

Solution

First we build the joint distribution. Since $\mathrm{P}(U = u,\, V = v) = \mathrm{P}(U = u)\,\mathrm{P}(V = v \mid U = u)$ and $\mathrm{P}(U = u) = \frac{1}{3}$ for each u,

	$U = 1$	$U = 0$	$U = -1$
$V = 1$	$\frac{1}{9}$	$\frac{1}{6}$	$\frac{2}{9}$
$V = -1$	$\frac{2}{9}$	$\frac{1}{6}$	$\frac{1}{9}$

The six entries add to 1, as they should.

Part (i)

The roots of $x^2 + Ux + V = 0$ are both real exactly when the discriminant is non-negative: $U^2 - 4V \geq 0$.

- If $V = -1$, the discriminant is $U^2 + 4 > 0$, so the roots are real (and distinct) whatever U is.
- If $V = 1$, the discriminant is $U^2 - 4 \leq 1 - 4 < 0$, since $U^2 \leq 1$. So the roots are never real.

Hence both roots are real if and only if $V = -1$, and

$$\mathrm{P}(\text{both roots real}) = \mathrm{P}(V = -1) = \tfrac{2}{9} + \tfrac{1}{6} + \tfrac{1}{9} = \tfrac{4+3+2}{18} = \tfrac{1}{2},$$

as required.

Part (ii)

By part (i), conditioning on "both roots real" is the same as conditioning on $V = -1$. Dividing the bottom row of the table by $\mathrm{P}(V = -1) = \frac{1}{2}$ gives

$$\mathrm{P}(U = 1 \mid V = -1) = \tfrac{4}{9}, \qquad \mathrm{P}(U = 0 \mid V = -1) = \tfrac{1}{3}, \qquad \mathrm{P}(U = -1 \mid V = -1) = \tfrac{2}{9}.$$

Ask yourself Why can't we just use $\mathrm{P}(U = u) = \frac{1}{3}$ here?

Because U and V are not independent: knowing that $V = -1$ makes $U = 1$ more likely (since $V = -1$ has probability $\frac{2}{3}$ when $U = 1$). The expectation we want is conditional, so the weights must be the conditional probabilities.

With $V = -1$ the equation is $x^2 + Ux - 1 = 0$, whose larger root is

$$R = \frac{-U + \sqrt{U^2 + 4}}{2}.$$

So $R = \frac{1}{2}(\sqrt{5} - 1)$ when $U = 1$, $R = 1$ when $U = 0$, and $R = \frac{1}{2}(\sqrt{5} + 1)$ when $U = -1$. Therefore

$$\begin{aligned} \mathrm{E}(R \mid \text{roots real}) &= \tfrac{4}{9} \cdot \frac{\sqrt{5} - 1}{2} + \tfrac{1}{3} \cdot 1 + \tfrac{2}{9} \cdot \frac{\sqrt{5} + 1}{2} \\ &= \frac{4\sqrt{5} - 4 + 6 + 2\sqrt{5} + 2}{18} \\ &= \frac{6\sqrt{5} + 4}{18} = \frac{3\sqrt{5} + 2}{9}. \end{aligned}$$

Part (iii)

The constant term is U, which suggests a factor $(x + U)$. Indeed

$$\begin{aligned} (x + U)(x^2 - 2Vx + 1) &= x^3 - 2Vx^2 + x + Ux^2 - 2UVx + U \\ &= x^3 + (U - 2V)x^2 + (1 - 2UV)x + U, \end{aligned}$$

which is exactly the given cubic. So the roots are $x = -U$ together with the roots of $x^2 - 2Vx + 1 = 0$. That quadratic has discriminant $4V^2 - 4 = 0$, because $V = \pm 1$ gives $V^2 = 1$. So it has a repeated root $x = V$, and the cubic is

$$(x + U)(x - V)^2 = 0,$$

with roots $-U$, V, V.

All three roots are positive if and only if $-U > 0$ and $V > 0$, that is, $U = -1$ and $V = 1$. (When $U = 0$ one root is 0, which is not positive.) From the table,

$$\mathrm{P}(\text{all roots positive}) = \mathrm{P}(U = -1,\ V = 1) = \tfrac{1}{3} \cdot \tfrac{2}{3} = \tfrac{2}{9}.$$

Checks

- The conditional probabilities $\frac{4}{9}, \frac{1}{3}, \frac{2}{9}$ add to 1.
- The answer to (ii) is about 0.967. It must lie between the smallest possible larger root, $\frac{1}{2}(\sqrt{5}-1) \approx 0.618$, and the largest, $\frac{1}{2}(\sqrt{5}+1) \approx 1.618$, and it should sit below the middle value 1, because the smallest root carries the biggest weight. It does.
- In (iii), try $U = -1$, $V = 1$: the cubic becomes $x^3 - 3x^2 + 3x - 1 = (x-1)^3$, with roots $1, 1, 1$, all positive. Try $U = 1$, $V = 1$: it becomes $x^3 - x^2 - x + 1 = (x+1)(x-1)^2$, which has the negative root -1.
- We confirmed all three answers by exact enumeration over the six cases with fractions, and by finding the cubic's roots numerically in each case.

Common slips

- Using the unconditional weights $\frac{1}{3}$ in (ii), which gives $\frac{1}{3}(\sqrt{5}+1) \approx 1.079$ instead.
- Counting $U = 0$ in (iii): a root of 0 is not positive.
- Trying to analyse the cubic with discriminants or sign changes instead of looking for the factorisation.

Beyond STEP

The heart of this question is conditional expectation, $\mathrm{E}(X \mid A)$, computed by reweighting with conditional probabilities. At university this idea grows into the conditional expectation $\mathrm{E}(X \mid Y)$ as a random variable in its own right, and the tower law $\mathrm{E}\big(\mathrm{E}(X \mid Y)\big) = \mathrm{E}(X)$, which makes many hard expectations easy. Grimmett and Welsh, *Probability: An Introduction*, develops this clearly.

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